# Algebra 2 — Unit 1: Linear Functions

**Practice Test — Answer Key**

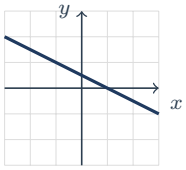
**Instructions:** Show all work. Write every line in $y = mx + b$ form when asked. No notes or calculator unless your teacher says so. **Total: 100 pts**

**Keep in Mind**

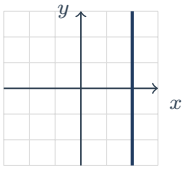
**This is a practice test.** It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers differ. Work every problem with no notes, then check your answers against the key.

## Part A — Types of Slope (8 pts)

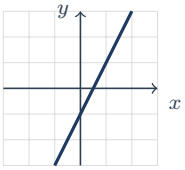
For each line, write whether its slope is **positive**, **negative**, **zero**, or **undefined**.



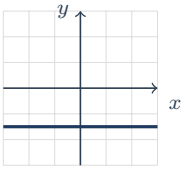

1. negative




2. undefined




3. positive




4. zero

No graph this time — decide from the points or the equation.

5. The line through (3, 2) and (−1, 2): **zero**

6. The line $y = 5x - 1$: **positive**

7. The line $x = -4$: **undefined**

8. The line through (0, 6) and (2, 1): **negative**

## Part B — Writing Equations of Lines (20 pts)

Write each line in slope-intercept form, $y = mx + b$. Show how you found $m$ and $b$; for 11 and 12, start in point-slope form, $y - y_1 = m(x - x_1)$.

9. The line through (1, 5) and (3, 9).

$$m = \frac{9 - 5}{3 - 1} = \frac{4}{2} = 2$$

$$5 = 2(1) + b \quad \Rightarrow \quad b = 3$$

Equation: $y = 2x + 3$

10. The line through (−4, 5) and (2, −4).

$$m = \frac{-4-5}{2-(-4)} = \frac{-9}{6} = -\frac{3}{2}$$

$$5 = -\frac{3}{2}(-4) + b = 6 + b \quad \Rightarrow \quad b = -1$$

Equation: $y = -\frac{3}{2}x - 1$

11. The line with slope 2 that passes through (3, 4).

$$y - 4 = 2(x - 3)$$
$$y - 4 = 2x - 6$$
$$y = 2x - 2$$

Equation: $y = 2x - 2$

12. The line with slope $-\frac{1}{3}$ that passes through (6, 1).

$$y - 1 = -\frac{1}{3}(x - 6)$$
$$y - 1 = -\frac{1}{3}x + 2$$
$$y = -\frac{1}{3}x + 3$$

Equation: $y = -\frac{1}{3}x + 3$

## Part C — Piecewise Functions (20 pts)

13. **Evaluate.** (8 pts) $f(x) = \begin{cases} 3x+2, & x < -1 \\ -1, & -1 \le x \le 1 \\ 2x-5, & x > 1 \end{cases}$

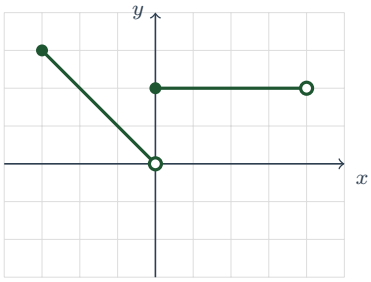
$f(-2) =$ $-4$ $\quad f(-1) =$ $-1$ $\quad f(1) =$ $-1$ $\quad f(4) =$ $3$

14. **Domain.** (12 pts) Give the domain of each function as an inequality or in interval notation.

(a) $g(x) = \begin{cases} x-2, & x < 3 \\ 7, & 3 \le x < 6 \end{cases}$ Domain: $x < 6$, $(-\infty, 6)$

(b) $h(x) = \begin{cases} x+1, & -2 \le x \le 0 \\ 4-x, & 0 < x \le 5 \end{cases}$ Domain: $-2 \le x \le 5$, $[-2, 5]$

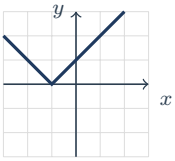
(c) The function $k$ graphed below.



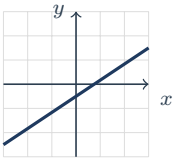

Domain: $-3 \le x < 4$, $[-3, 4)$

## Part D — The Absolute Value Function (28 pts)

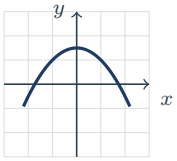
15. (3 pts) Circle the letter of the graph of an **absolute value** function. **Letter:** A



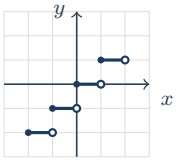

→ **(A)**




(B)




(C)




(D)

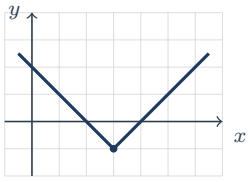
16. (3 pts) Which equation is an **absolute value** function? **Letter:** C

(A) $y = x^2 - 3$ (B) $y = 2x + 5$ → **(C)** $y = |x + 1| - 3$ (D) $y = \lfloor x \rfloor + 1$

17. (15 pts) Each function is a transformation of the parent $y = |x|$. Fill in the table.

| Function | Vertex | Left/right shift | Up/down shift | Opens |
|---|---|---|---|---|
| $g(x) = \lvert x + 2 \rvert + 3$ | $(-2, 3)$ | left 2 | up 3 | up |
| $h(x) = \lvert x - 5 \rvert - 4$ | $(5, -4)$ | right 5 | down 4 | up |
| $k(x) = -\lvert x \rvert - 2$ | $(0, -2)$ | none | down 2 | down |

18. (4 pts) Which equation matches the graph? **Letter:** C




(A) $y = |x + 3| - 1$

(B) $y = |x - 3| + 1$

→ **(C)** $y = |x - 3| - 1$

(D) $y = -|x - 3| - 1$

19. (3 pts) Compared with the parent $y = |x|$, the graph of $y = \frac{1}{2}|x|$ is **Letter:** B

(A) narrower → **(B)** wider (C) shifted down 2 (D) shifted left 2

## Part E — Absolute Value Equations & Inequalities (24 pts)

Solve. Show both cases.

20. (8 pts) $|2x + 1| = 9$

$$2x + 1 = 9 \quad \text{or} \quad 2x + 1 = -9$$
$$2x = 8 \quad \text{or} \quad 2x = -10$$

Solution: $x = 4$ **or** $x = -5$

21. (8 pts) $|x - 3| \leq 2$

$$-2 \leq x - 3 \leq 2$$
$$1 \leq x \leq 5$$

Solution: $1 \leq x \leq 5$

22. (8 pts) $|x + 4| > 6$

$$x + 4 > 6 \quad \text{or} \quad x + 4 < -6$$
$$x > 2 \quad \text{or} \quad x < -10$$

Solution: $x > 2$ **or** $x < -10$